GRADE 3 · INTRO PYTHON TURTLE

Meet Python Turtle

Ontario Math C3.1 — repeating events, now in typed code

Name: .... Date: ....

★ I can read a typed loop that makes Bit move again and again.

Same loop — now typed

You already used a repeat block to make Bit do something many times. Python Turtle does the SAME thing — but you TYPE it instead of dragging blocks.

Here is the repeat you know, and the same idea written in Python. Bit is the turtle that draws.

The block you know

repeat 3

  move 20

Move 20, three times.

The same thing in Python

PYTHON

```
for step in range(3):
    bit.forward(20)
```

OUTPUT

```
Bit moves forward 20, three times → 60 steps.
```

range(3) → 3 hops → 60 steps

range(3) means do the inside 3 times — Bit makes 3 hops of 20.

1 Read it

Look at the Python code. range(3) repeats the inside 3 times. How many times does Bit move forward? ______ Each hop is 20, so how far in TOTAL? ______

2 Predict, then run

You change range(3) to range(5). PREDICT: how many hops, and how far in total? Then run it to check.

3 Write your own

Make Bit move forward 10, FOUR times. Fill in the two numbers: for step in range(______): bit.forward(______)

4 Challenge

Bit's code is for step in range(4): bit.forward(25). How far does Bit travel in total? Show how you worked it out.

Big idea

range(n) in a Python loop is just like repeat n in blocks — it does the inside n times. Typed code can repeat, exactly like blocks can.